

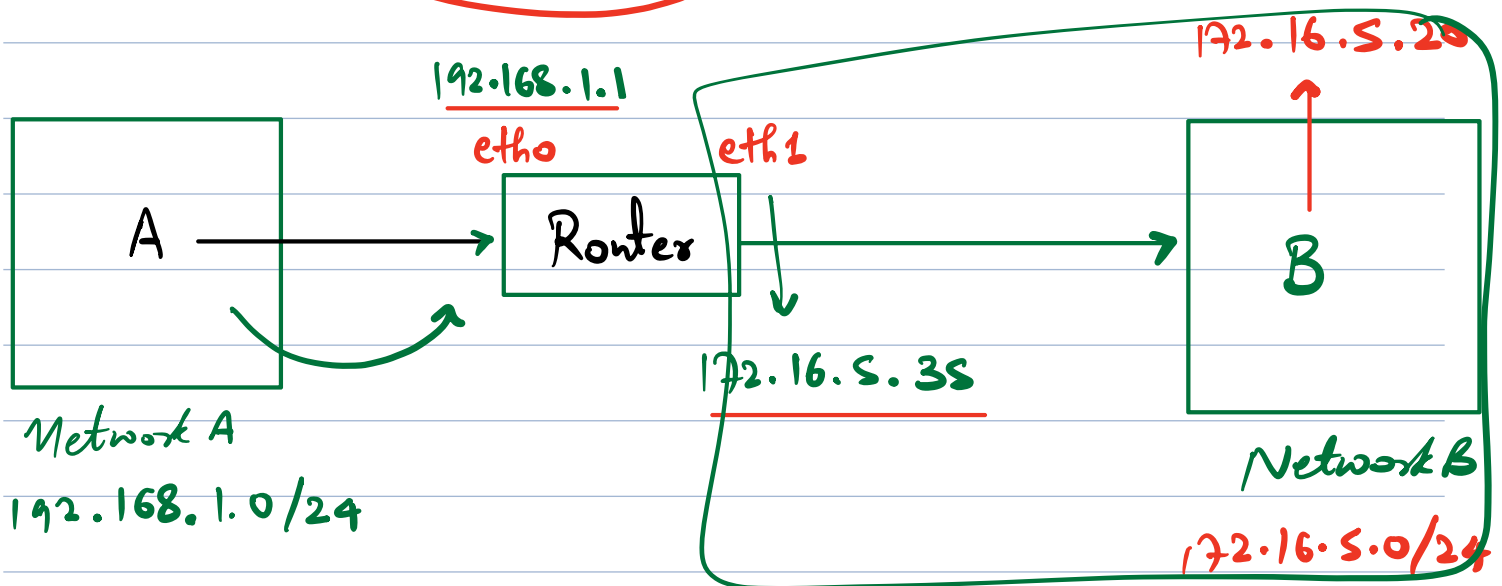
A $\Rightarrow$ IP: 192.168.1.10
SM: 255.255.255.0
Prefix: /24
DG: 192.168.1.1

A $\longrightarrow$ (192.168.1.50)

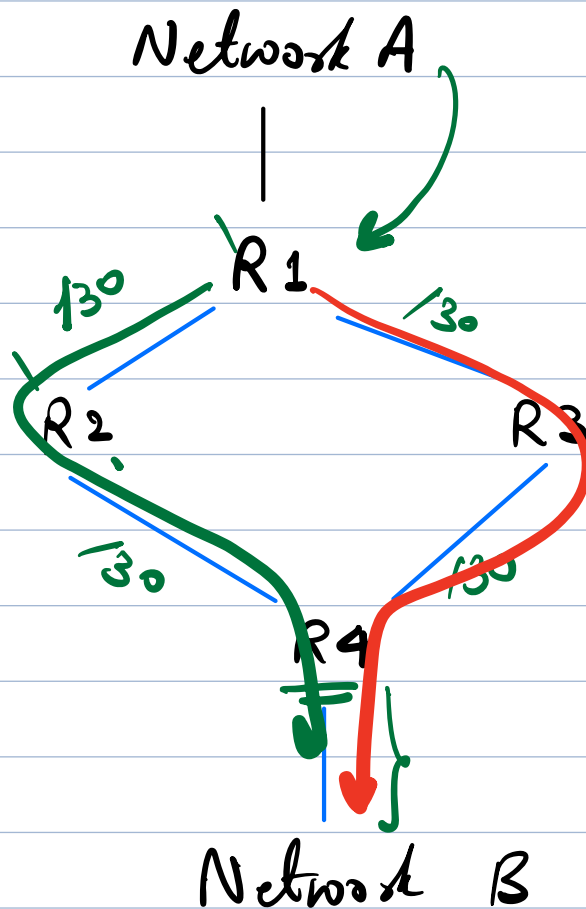
192.168.1.0/24

A $\longrightarrow$ (172.16.5.20) Outside Network
192.168.1.10

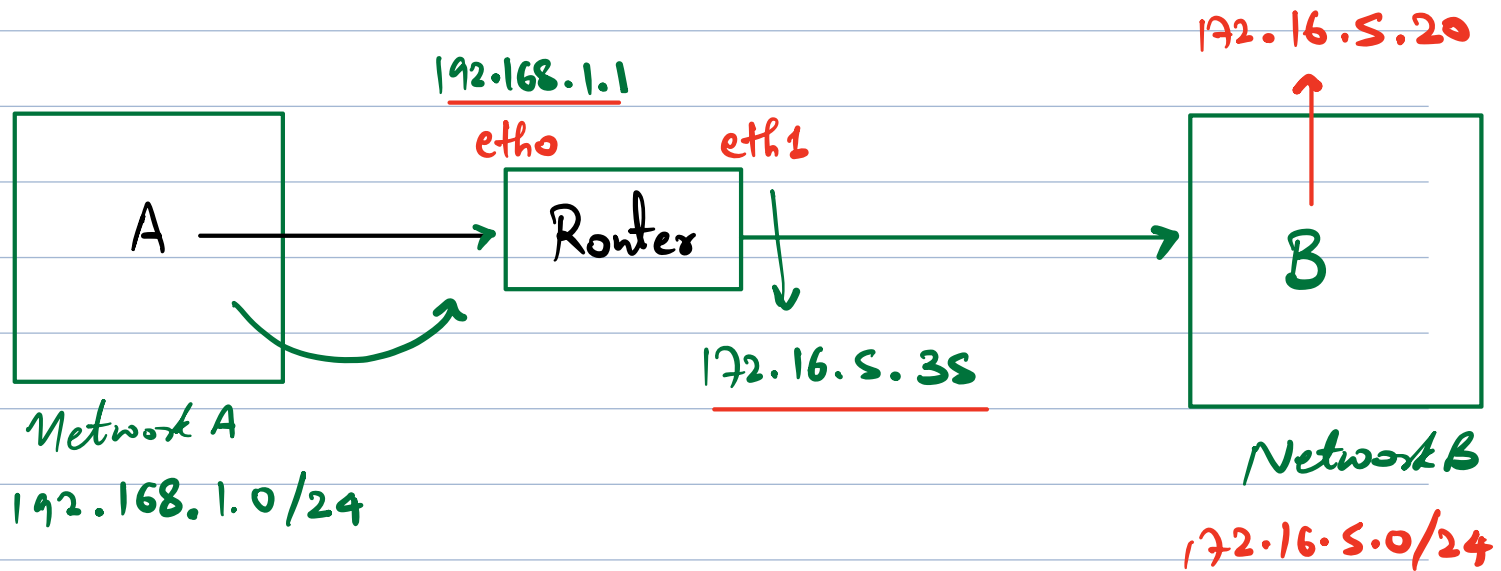
A $\longrightarrow$ Gateway Router ??



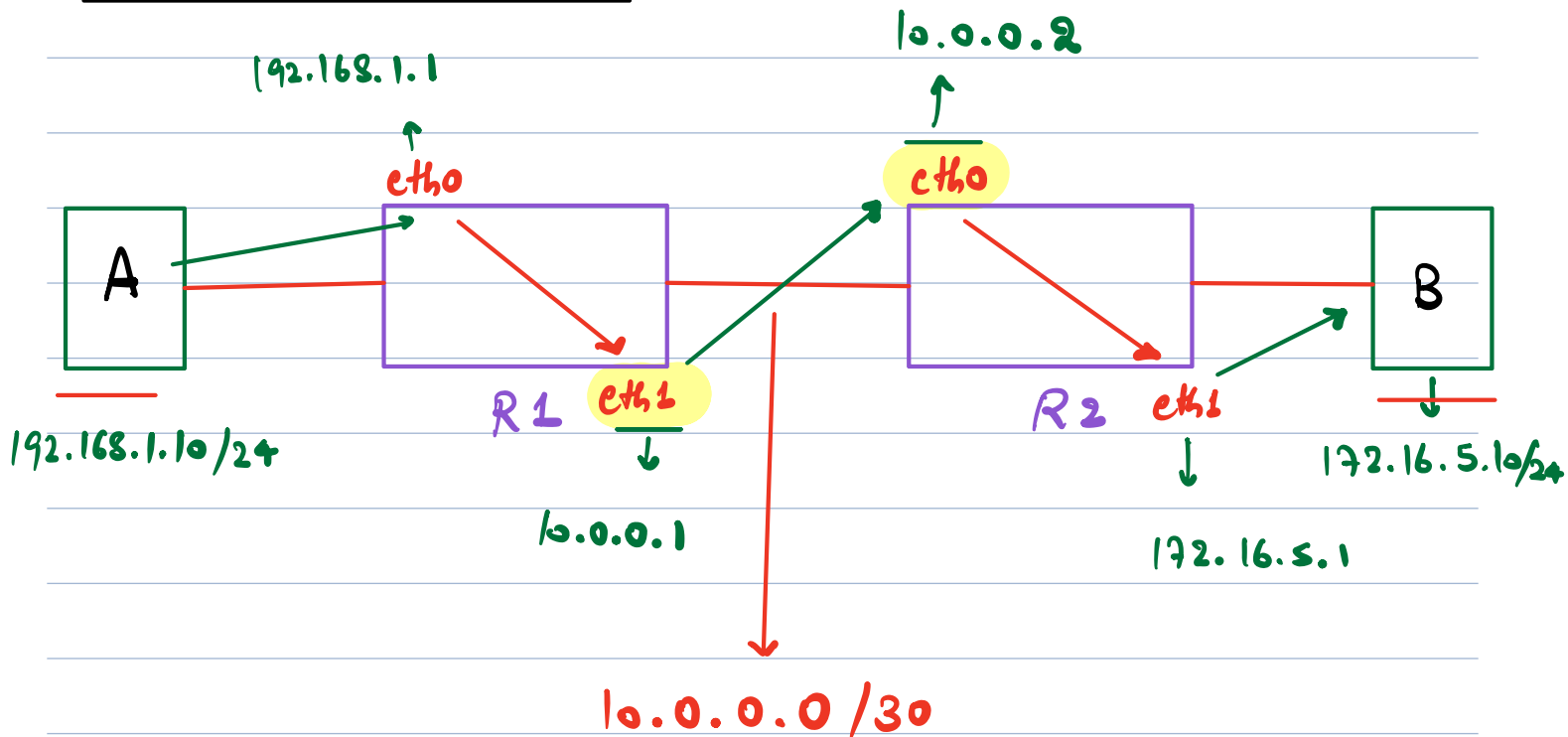
NH $\Rightarrow$ Src - 192.168.1.0
dest $\rightarrow$ 172.16.5.20



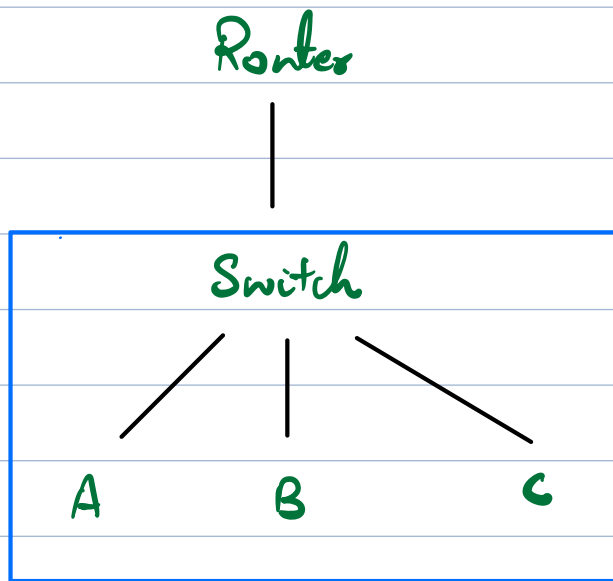
Router Interfaces $\neq$ Directly
Connected Networks



Remote Network

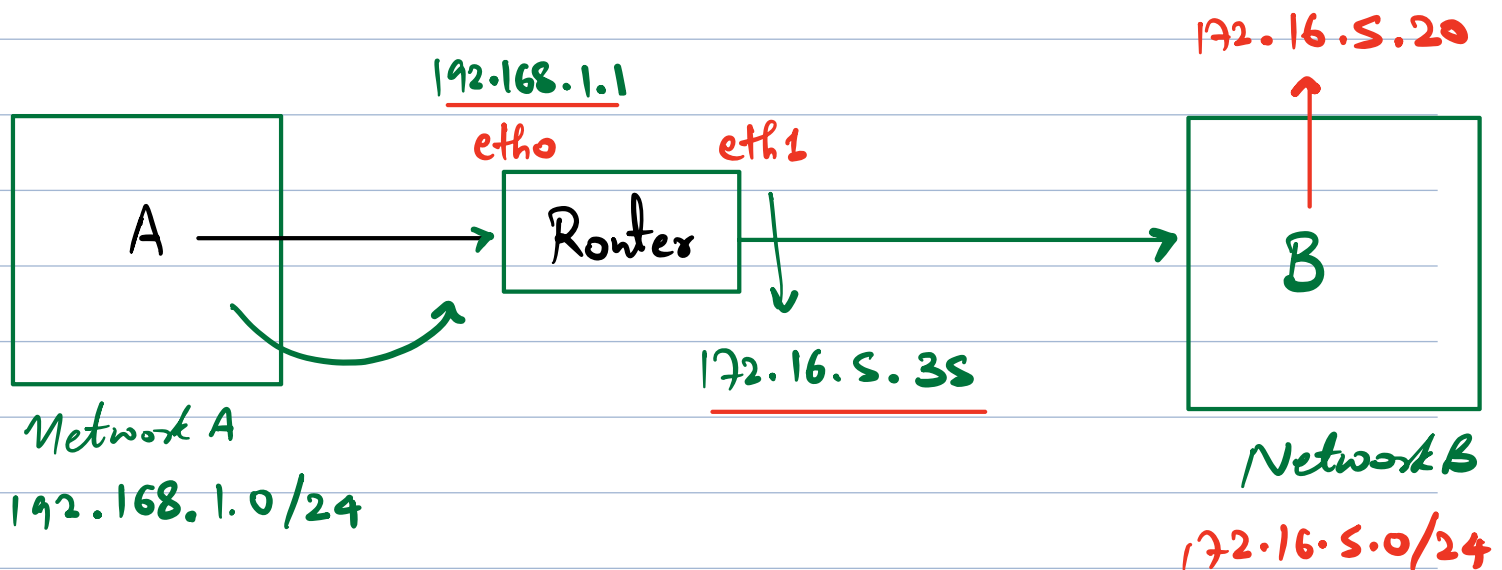


src → 192.168.1.10
 dest → 172.16.5.10



* Directly connected router appears automatically

Routing Table



Dest IP	Next Hop	interface
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192.168.1.0/24

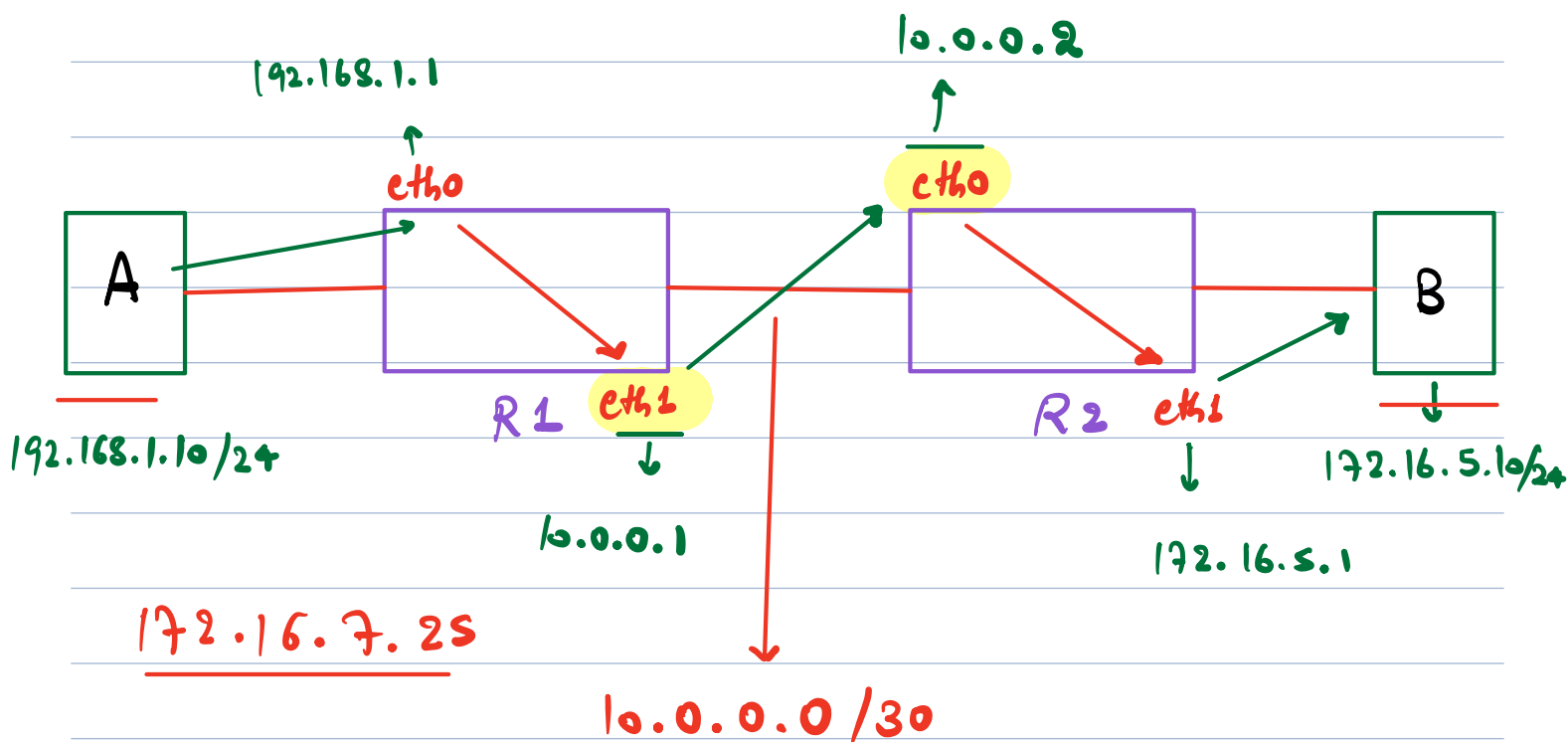
Direct

eth0

172.16.5.0/24

Direct

eth1



R1

Dest IP	Next Hop	interface
---------	----------	-----------

192.168.1.0/24

Direct

eth0

10.0.0.0/30

Direct

eth1

172.16.5.0/24

10.0.0.2

eth1

172.16.1.0/24

via 10.0.0.2

through eth1

R2

Dest IP	Next Hop	interface
172.16.5.0/24	Direct	eth 1
10.0.0.0/30	Direct	eth 0
192.168.1.0/24	10.0.0.1	eth 0

Routing Table

Destination Prefix

Next Hop

Outgoing Interface

! Where do Routing Table Entries come from ??

1) Connected Routes

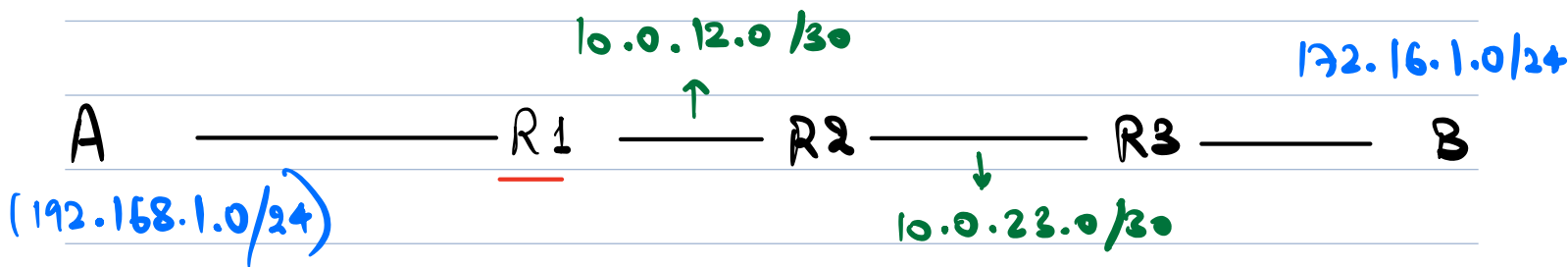
→ Automatically derived from active configured interfaces

2) Static Routes

→ Configured manually.

3) Dynamic Routes

⇒ Learned using routing protocols.



Dest	Next Hop	Interface
192.168.1.0/24	Direct	eth0
10.0.12.0/30	Direct	eth1
10.0.23.0/30	10.0.12.2	eth1
172.16.1.0/24	10.0.12.2	eth1

Dest	Next	Interface
172.16.0.0/16	10.0.0.2	eth1
172.16.20.0/24	10.0.0.6	eth2
0.0.0.0/0	10.0.0.2	eth1

172.16.20.50

1) 172.16.0.0/16 ✓

2) 172.16.20.0/24 ✓

3) 0.0.0.0/0 ✓

Longest Prefix Match

Consider the following routing table:

Destination Prefix	Next Hop	Interface
<u>10</u> .0.0.0/8	192.168.1.1	eth0
<u>10</u> . <u>10</u> .0.0/16	192.168.2.1	eth1
<u>10</u> . <u>10</u> . <u>20</u> .0/24	192.168.3.1	eth2
<u>10</u> . <u>10</u> . <u>20</u> . <u>128</u> /25	192.168.4.1	eth3
0.0.0.0/0	192.168.5.1	eth4

Now suppose a packet arrives with:

10.10.20.150 → 10010110

① 0 0 0 0 0 0 0 0 → 128
1 1 1 1 1 1 1 1 → 255
129
254

More network bits



Smaller destination range



More specific route

First:

Which route describes this destination most specifically?

Then:

If routing logic has multiple equivalent-prefix candidates,
other criteria may be used.



Dest Prefix

Next Hop

172.16.0.0/12 ✓
172.16.0.0/16 ✓

R1

R2

172.16.10.0 /24 ✓
172.16.10.64 /26 ✗
0.0.0.0 /0 ✓

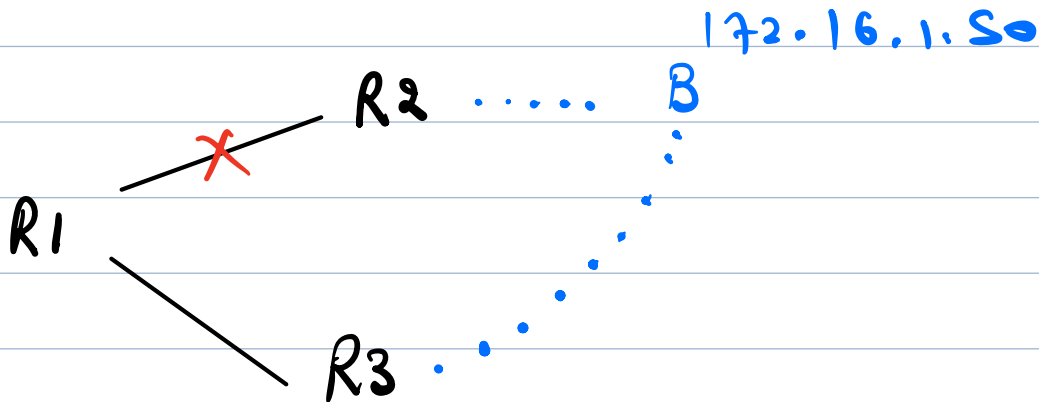
R3 ✓
R4
R5

172.16.10.200 → (11)001000

(01000000)

Static Routing

A **static route** is a route manually configured by a network administrator.



Dest Prefix

Next Hop

172.16.1.0/24

R2

! When is Static Routing Useful??

1) The Network is Small.

2) The Path is Predictable

A — R1 ——— R2 ——— B

3) Explicit Control.

4) Special or Backup Routes.

Static Routing	Dynamic Routing
Manually configured	Learned using routing protocols
Simple for small networks	Better for larger networks
Predictable	Adapts to network changes
Low protocol overhead	Requires routing communication
Manual updates	Routes can update automatically

```
ip route add 172.16.1.0/24 via 10.0.0.2
```

```
ip route add 172.16.1.0/24 via 10.0.0.2 dev eth1
```

```
sudo ip neigh add 10.0.0.2 lladdr aa:bb:cc:dd:ee:ff dev eth1
```